
Table of Contents

.....	1
Complex Step	1
Analytical	1
Plotting	1
Functions	2

```
clc; clear all; close all;
```

```
t_vals = 0:.01:3;
```

```
P = sin(t_vals);
```

```
c = exp(t_vals);
```

```
theta = [P; c];
```

```
h = 1e-12;
```

Complex Step

```
z_p_P = complexStep(theta, 1, h, t_vals);
```

```
z_p_c = complexStep(theta, 2, h, t_vals);
```

```
z_ps = [z_p_P, z_p_c];
```

```
names = ['P', 'c'];
```

Analytical

```
syms P c t
```

```
t0 = 0;
```

```
z = (P/c^2) * (exp(-c*(t-t0)) - 1 + c*(t-t0));
```

```
% dz/dP
```

```
dz_dP = simplify(diff(z, P));
```

```
%dz/dc
```

```
dz_dc = simplify(diff(z, c));
```

```
z_ = zeros(2, numel(t_vals));
```

```
for i = 1:numel(t_vals)
```

```
    z_(1, i) = subs(dz_dP, [P, c, t], [theta(1, i), theta(2, i), t_vals(i)]);
```

```
    z_(2, i) = subs(dz_dc, [P, c, t], [theta(1, i), theta(2, i), t_vals(i)]);
```

```
end
```

Plotting

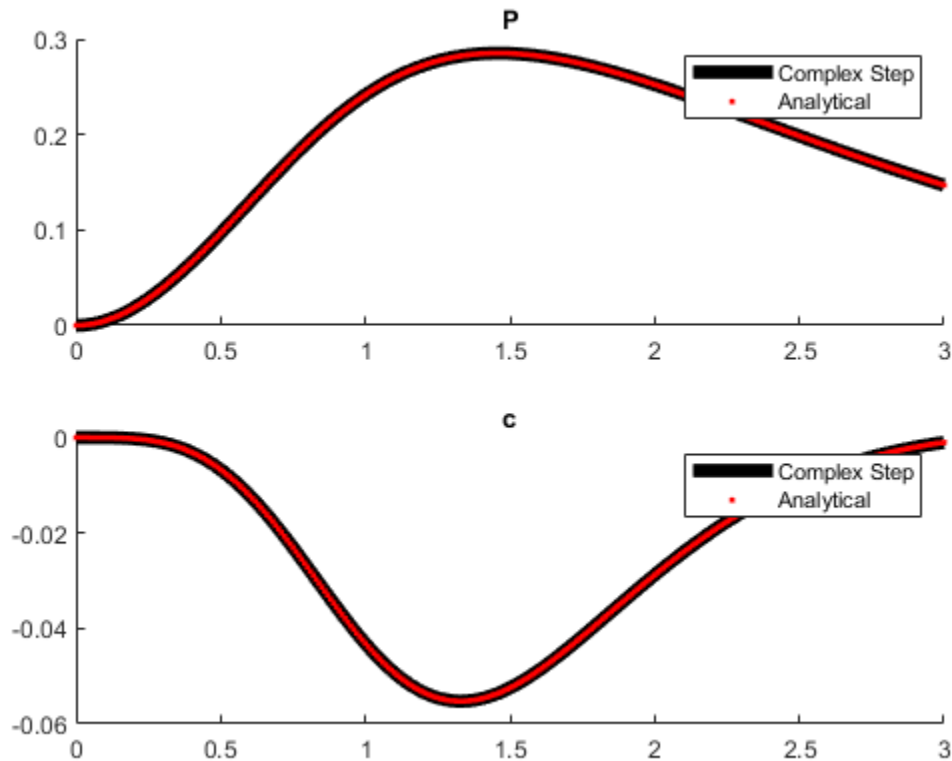
```
figure()
```

```
for i = 1:2
```

```

subplot(2, 1, i)
hold on
plot(t_vals, z_ps(:, i), 'k', 'DisplayName', 'Complex Step',
'LineWidth', 5);
plot(t_vals, z_(i,:), 'r.', 'DisplayName', 'Analytical', 'LineWidth', 1);
hold off
legend()
title(names(i));
end

```



Functions

```

function z_p = complexStep(thetas, pos, h, t_vals)
    z_p = zeros(numel(t_vals), 1);
    thetas(pos, :) = thetas(pos, :) + 1i * h;

    for i = 1:numel(t_vals)
        t_val = t_vals(i);
        z = xfr(thetas(:, i), t_val);
        z_p(i) = imag(z) / h;
    end

end

function z = xfr(theta, t)
    t0 = 0;

```

```
P = theta(1);  
c = theta(2);  
  
z = (P/c^2) * (exp(-c*(t-t0)) - 1 + c*(t-t0));  
  
end
```

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